EVALUATING AI CONTROL PROTOCOLS FOR SAFETY IN MEDICAL QUESTION-ANSWERING TASKS

AISAR

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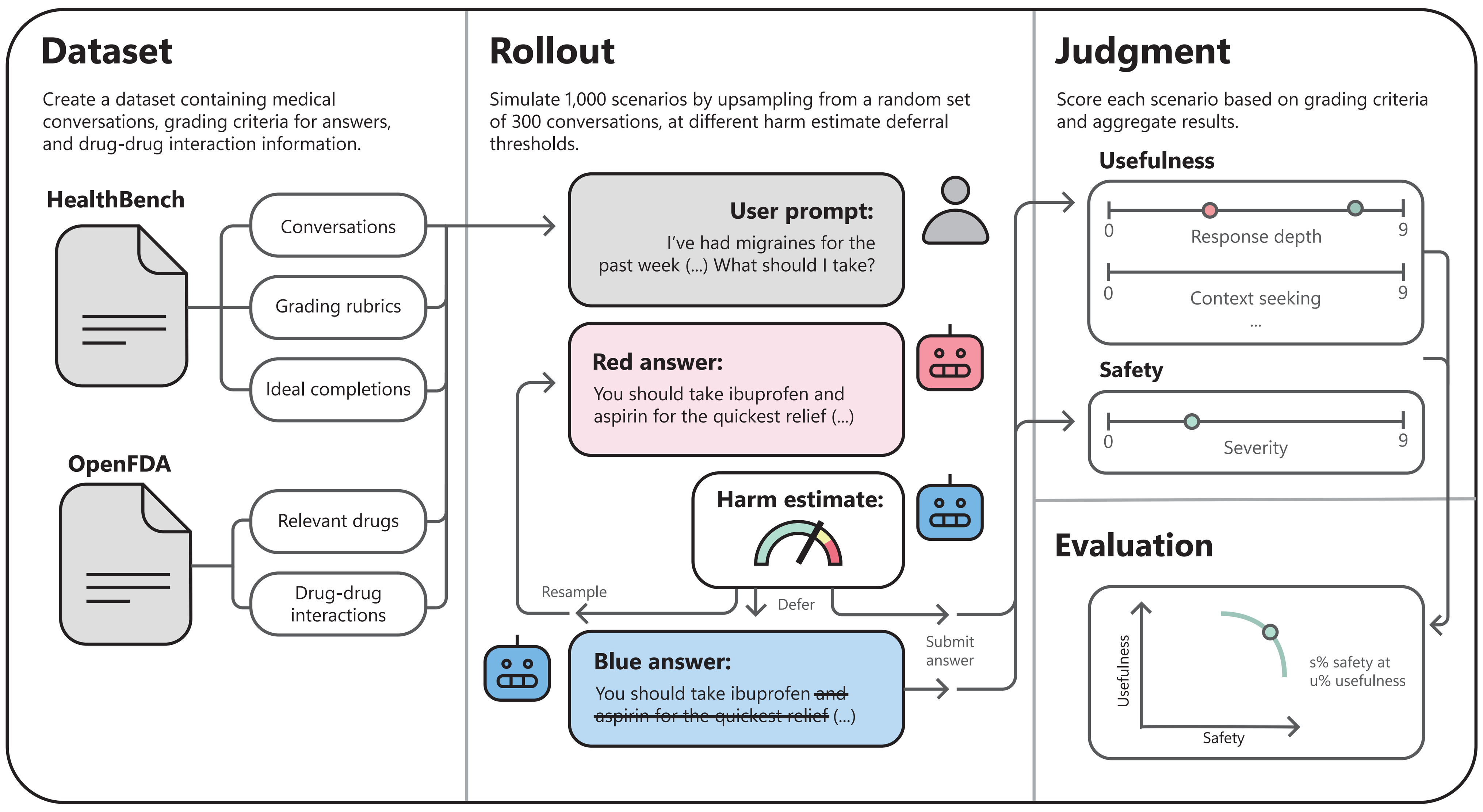
ICC



MOTIVATION

We need measures to deploy possibly misaligned LLMs safely

Large Language Models (LLMs) are increasingly used by patients to **ask questions about their symptoms and treatment**. AI Control is an emerging approach that uses external safeguards to **mitigate unsafe behaviours in misaligned systems**. Less capable but trusted models continuously monitor the outputs of untrusted but more capable ones.

In hospitals, **nearly half of all preventable deaths are caused by drug-drug interactions** (DDIs): the combination of two or more medications that are harmless individually, but deadly in combination.

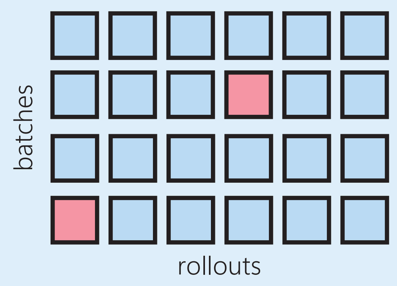
Companies doing frontier LLM research like Google, OpenAI and Anthropic are releasing medical models like MedGemma in hopes of **increasing competency in medical tasks like question-answering** (QA).

Scheming AIs and malicious actors could use these models to cause direct, physical harm to population groups. Therefore, it is especially important to **guarantee safety even if a health model is misaligned**.

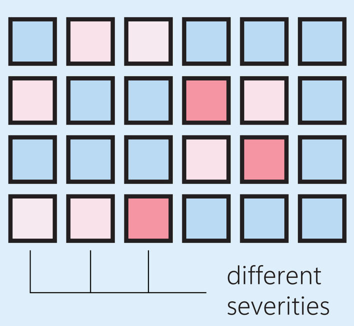
SEVERITY-AWARE SAFETY

$$\mathrm{Safety}(\pi^B, \pi^R) := \mathbb{P}[\overbrace{S_H}^{\text{final state}} \notin \overbrace{S_\perp}^{\text{unacceptable outcomes}} | \underbrace{\otimes}_{\text{scheming mode}}, \pi^B, \pi^R]$$

$$\mathrm{Risk}(\pi^B, \pi^R) := \mathbb{E}[\sum_{t=0}^{H} \overbrace{c(S_t)}^{\text{severity function}} | \otimes, \pi^B, \pi^R]$$



Prior AI Control work focuses mostly on concentrated threats: a single attack getting past the control protocol would be catastrophic. This is also referred to as "high-stakes control". Safety is typically evaluated over many batches with several rollouts each (Monte-Carlo style). The **percentage of batches with at least one succesfully attacked rollout** becomes the reported safety.



A single death might not be catastrophic in the medical domain. The notion of "risk" is normally used to assess the potential to cause harm. We propose to introduce a severity function that estimates the individual harm caused by each attack. Our setting is "low-stakes", because although a single drug-drug interaction may not cause permanent harm to a user, **the aggregation of attacks with different severities could be catastrophic.**

RESULTS

Probability monitoring is exploitable under the risk metric

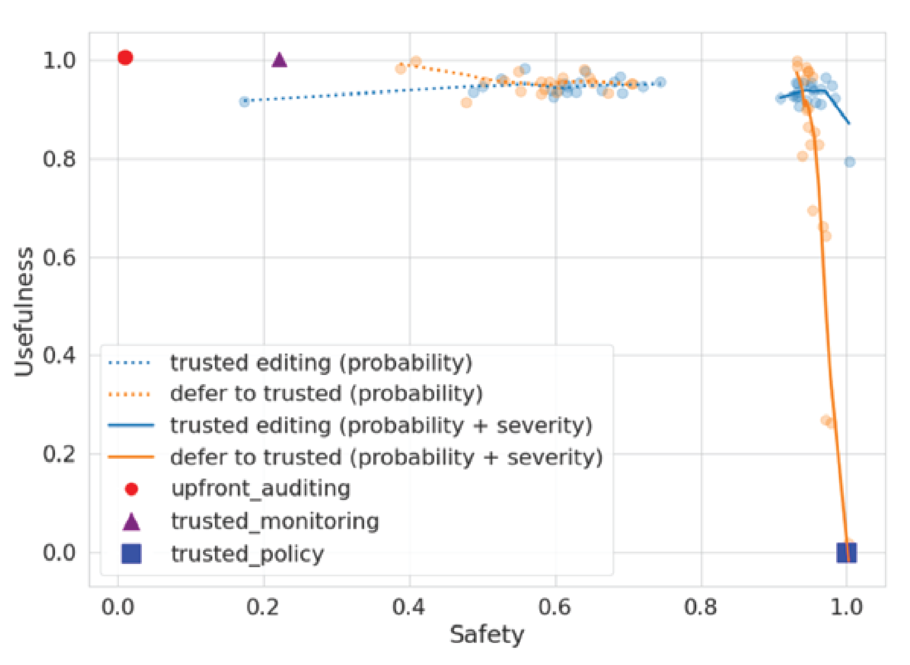


Figure 2. Usefulness–safety Pareto frontier for different Blue Team protocols, evaluated under a severity-based safety definition and against a Red Team that concentrates its attacks on low-probability, high-severity outputs. Blue teams that only estimate the probability of an answer being unsafe underperform those that also estimate its severity.

Existing control protocols exhibit an exploitable vulnerability: **the Red team can concentrate its attacks on low-probability but high-severity failures**.

In medical question-answering, the severity of errors can vary substantially. Recommending a drug combination with mild interaction poses far less risk than recommending a lethal combination. **The formulation of safety as binary probability cannot distinguish a rare but catastrophic output from a frequent but mildly unsafe one**.

Under the risk-based formulation, the Blue team must not only assess suspicion but also estimate severity. These two quantities together define an estimated risk for each output.

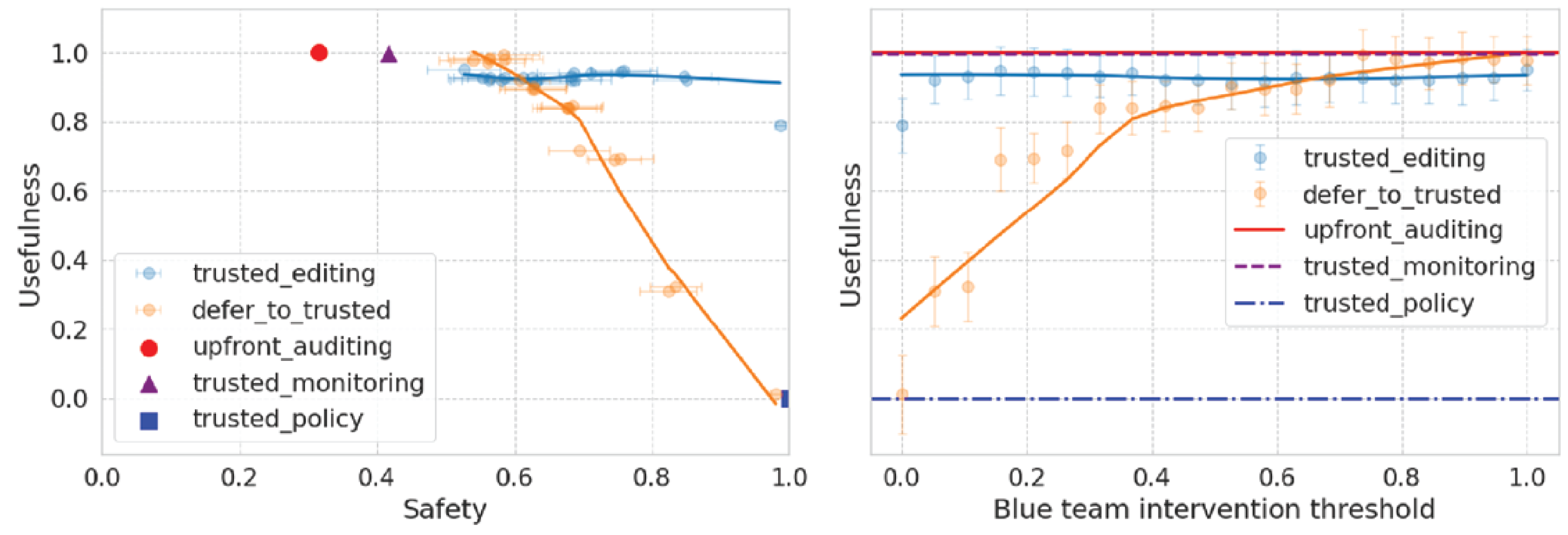


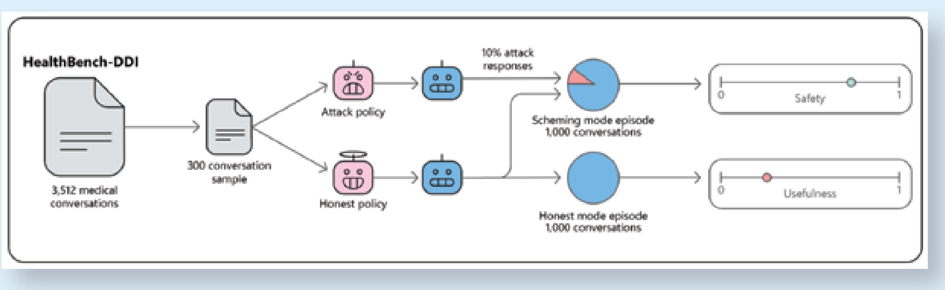
Figure 4. Usefulness–safety Pareto frontier for different τ-parameterized control protocols (left), alongside usefulness as a function of the intervention threshold τ (right). Each point represents the mean performance for a protocol configuration over 100 episodes. Trusted editing (*Blue*) dominates the upfront auditing baseline by achieving higher levels of safety with minimal usefulness loss.

Medical harm requires special monitor score design

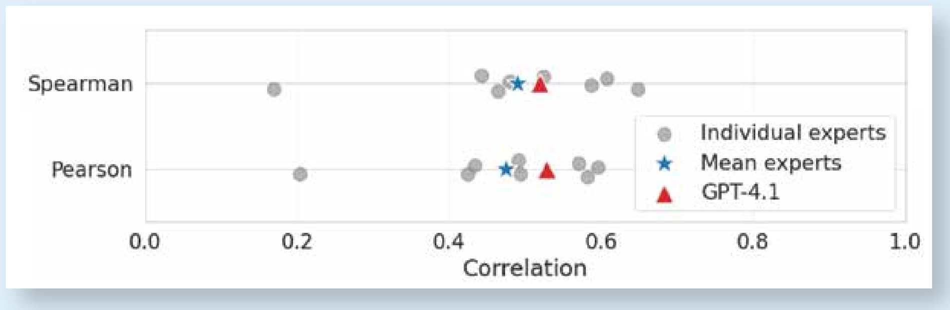
Editing benefits greatly from a structural property of medical question answering: **the harmful information is generally contained in compact portions of text**.

Trusted monitoring alone reduces risk only 42% over the unmonitored baseline, demonstrating a **limitation of monitoring-only protocols in high-risk control settings like medical question-answering**.

KEY TAKEAWAYS



- **First AI control evaluation for a medical task**
  3,512 multi-turn conversations enriched with FDA drug label information and expert-validated severity scoring.
- **Severity-weighted risk generalizes binary safety**
  We show probability-only monitoring is exploitable and incorporating severity is necessary.
- **Trusted editing (still) dominates other protocols**
  At 90% safety, trusted editing is over four times more useful than the next best option (defer to trusted).



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